Dual-Collaborative Fusion Model for Multispectral and Panchromatic Image Fusion

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***Abstract*— The aim of multispectral (MS) and panchromatic (PAN) image fusion is to obtain an MS image that has high resolution in both spectral and spatial domains. During the fusion process, there are two important issues, i.e., spectral information preservation and spatial information enhancement. In this article, we propose a dual-collaborative fusion model that considers not only the spectral correlation collaboration but also the spatial-spectral collaboration. First, the features of PAN and MS images are extracted by a shared feature embedding network. Then, in order to enhance the spatial details, the PAN features are decomposed into four subbands, and the collaborative relationships among subbands are fully explored to refine the features. After the refinement of the subbands, the high-frequency components are directly taken as the inputs of the reconstruction network, while the low-frequency components are transformed by the guidance generation network to accomplish the spatial-spectral collaboration and also make preparations for the spectral adjustment. To explore the spectral correlation collaboration, a novel graph convolutional network is designed for the modulation of intraspectral relationships. Finally, the adjusted MS features are combined with the high-frequency components of PAN features to reconstruct the high-resolution MS image. Experimental results show that the proposed method outperforms traditional state-of-the-art pan-sharpening methods as well as the available deep learning-based ones.**

***Index Terms*— Deep learning (DL), dual-collaboration, image fusion, pan-sharpening, remote sensing.**

I. Introduction

With the development of the satellite imaging technique, a large amount of remote sensing data are provided by various optical earth observation satellites such as QuickBird, GeoEye, and WorldView, which are beneficial to many earth observation tasks. Most remote sensing applications, for example, land-cover classification [1], change detection [2], [3], and image segmentation [4], need an image with high resolution in both spatial and spectral domains, however, due to some physical limitations, a single sensor cannot provide such an image, which facilitates the development of remote sensing image fusion. Generally, remote sensing image fusion includes multispectral (MS) and panchromatic (PAN) image fusion, MS and hyperspectral (HS) image fusion [5], [6], synthetic aperture radar (SAR) and optical image fusion [7], [8]. Among them, the MS and PAN image fusion (MS+PAN) task together with the MS and HS image fusion (MS+HS) task is known as homogeneous image fusion tasks, whose goal is to integrate the abundant spatial information contained in PAN (MS for MS+HS task) and the comprehensive spectral information contained in MS (HS for MS+HS task) to obtain an image with high spatial and spectral resolution. The fusion process of MS and PAN images can be treated as the sharpening of MS images with the help of PAN images, and it is also known as pan-sharpening.

Traditional pan-sharpening methods are mainly divided into three groups [9], [10]: Component-substitution (CS)-based methods, multiresolution analysis (MRA)-based methods, and model-based optimization methods (MBO). CS-based methods also known as spectral transformation methods that always execute some kinds of spectral transformations on the MS image, and then one of the components is replaced partially or fully by the histogram-matched PAN image to obtain the high-resolution (HR) MS image. The popular CS-based methods include Gram–Schmidt (GS) method [11], [12], intensity-hue-saturation (IHS) method [13], [14], and principal component analysis (PCA) method [15]. The fusion results of CS-based methods often have fine spatial details, but the spectral information cannot be preserved well because the replaced component of MS inevitably has spectral information and the simple replacement will result in the spectral loss. MRA-based methods consider pan-sharpening from another perspective that the missing spatial details of MS images can be inferred from the high-frequency components of PAN images, and these high-frequency components are extracted by some MRA tools such as curvelet transform [16], contourlet transform [17], and support tensor transform [18]. Owing to the fact that MRA-based methods treat the fusion process as an extraction and injection operation of spatial details, they are also deemed to be spatial transformation methods. As opposed to CS-based methods, MRA-based methods

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always have strong spectral preservation ability, but if the spatial details extracted from PAN images are not matched with the missing parts of MS images, the aliasing effects and local dissimilarities will appear in the fusion products [18]. The MBO-based methods model the relationships between MS and PAN images and then build an energy function to reconstruct the HR MS image. Due to the fact that the reconstruction from the low-resolution (LR) image to the HR one is an ill-posed problem, the energy function of this type of methods is always combined with some reasonable regularization terms, such as sparse regularization [5], [19], low-rank recovery model [20], and variational model [21]. Although MBO-based methods make a tradeoff between spatial enhancement and spectral preservation, they always have high computational complexity.

Recently, many deep learning (DL)-based fusion methods [22]–[28] were proposed. To the best of our knowledge, the first DL-based method [22] utilized a modified sparse denoising autoencoder (MSDA) algorithm to learn the relationship between HR and LR PAN image patches and assumed that such a relationship is the same a that between HR/LR MS image patches, then the HR MS image could be reconstructed from the LR MS image patches and the trained model. The most similar work was proposed in [23], which drew the assumption from another perspective that the relationship between LR PAN and LR MS is the same as that between HR PAN and HR MS. The fusion results strongly rely on the rationality of the assumption, however, the imaging mechanisms of MS and PAN images are quite different, which will lead to distortions in the spatial and spectral domains. Benefit from the great success of DL technique in the field of single image super-resolution (SISR) [29]–[32], some convolutional neural network (CNN)-based fusion methods appeared. Motivated by the super-resolution convolutional neural network (SRCNN) [32], the pan-sharpening network (PNN) [24] was proposed intuitively. PNN had the same structure as SRCNN, but the authors took the nonlinear radiometric indices into account and also modified the inputs of the network to make it more appropriate to pan-sharpening. Later, Scarpa *et al.* [25] further explored a number of architectural and training variations to the PNN baseline and achieved a performance improvement through a lightweight network. Wei *et al.* [26] introduced the residual learning into the CNN structure to deepen the pan-sharpening network, and finally, the network had a relatively deeper structure and could extract more abstract features. However, both the spatial resolution and the spectral resolution are important to subsequent applications, and these networks paid more attention to spatial enhancement regardless of the spectral preservation. Yuan *et al.* [27] proposed a pan-sharpening network that has two branches of convolutional layers to extract spectral and spatial features based on the assumption that the spectral information and spatial details are contained in MS and PAN images, respectively. Meanwhile, they also utilized the multiscale feature extraction block to further locate the multiscale characteristics. Although this model extracted spatial and spectral information separately, there was no confirmation that the spectral correlation information could be extracted by the MS branch. Furthermore, due to the fact that the spatial details of SISR are inferred from the LR images rather than PAN images, these SISR-inspired fusion models tend to produce over-smooth results that lack some texture details. Zheng *et al.* [33] proposed an HS pan-sharpening model that first performed the guided filtering to obtain an initialized HS image, and then utilized the deep residual learning to regress the residuals. He *et al.* [34] designed a new detail injection based on the CNN framework for pan-sharpening. Similarly, detail-based deep Laplacian pansharpening (DDLPS) [35] also focused on the detailed injection, and made full use of the deep Laplacian pyramid framework. Although these methods obtained satisfactory results, they all stressed the spatial residuals and neglected the spectral information. In order to emphasize both the spatial details and the spectral information, Yang *et al.* [28] proposed the PanNet who achieved the spectral preservation by propagating the spectral information to the reconstructed image and accomplished the spatial enhancement by training the network parameters in high-pass filtering domain rather than image domain. PanNet could obtain excellent fusion results, but the propagation of spectral information was realized by the addition of original up-sampled MS image, and this simple addition inevitably introduced some undesirable low-frequency information, which would result in spectral distortions. Moreover, it treated the deep network as a black-box learner and did not take the collaborative relationships between spatial and spectral information into account, therefore, the exploration ability of spatial and spectral information is limited.

In this article, we make full use of the spatial information contained in PAN images together with the spectral information contained in MS images, and explore the spatial–spectral collaboration relationship in the fusion process to develop a dual-collaborative fusion (DCF) model for MS and PAN image fusion task. Inspired by the fact that different objects always have different spectral curves, and the differences of them are reflected in the relationships among MS bands, we stress the relationships in the fusion process. For a specific object, the response of each band is different, as shown in Fig. 1(a), and then if we use the differences of intensity values between any two bands to represent the distance between them, the constructed data are non-Euclidean because the distances of adjacent nodes are different, as illustrated in Fig. 1(b). Researchers usually utilize the graph structure to model the data with non-Euclidean structure, therefore, in this article, we model the relationships among different spectral bands by a series of graphs and use the graph convolutional network (GCN) [36]–[38] to adjust them. To emphasize the spatial details and avoid over-smooth fusion results, we utilize the stationary wavelet transform (SWT) [39] to decompose the PAN features into four subbands, and consider the relationships among them to refine the features. After the refinement, the features of low-frequency subband are exploited to obtain the spatial guidance and to accomplish the spatial–spectral collaboration, while the features of high-frequency subbands are combined with the adjusted MS features to obtain the fusion results. During this process, GCN is used to refine the spectral relationships among different bands in the HR domain. In general, the DCF model is composed of two branches,

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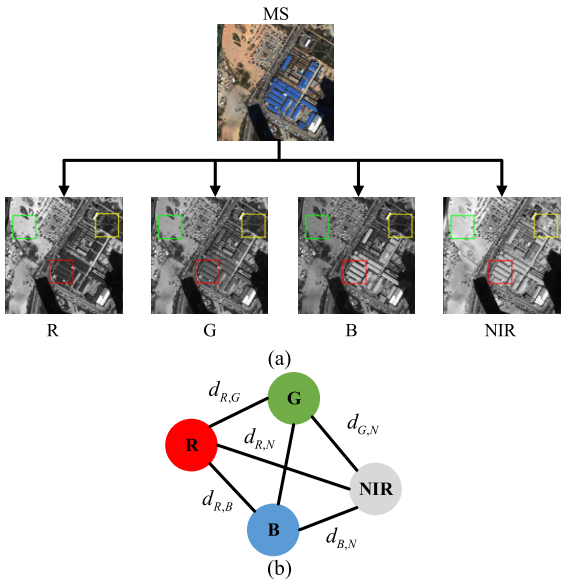


Fig. 1. Spectral composition of the MS image. (a) Spectral composition of a four-band MS image. (b) Spectral graph modeling.

i.e., the PAN branch and MS branch. The PAN branch consists of a feature embedding network (EbNet), a subband learning network, and a guidance generation (GG) network, whereas, the MS branch consists of a feature EbNet, the up-sampling operation and a spectral adjustment (SA) network. Finally, the fusion result is obtained through the reconstruction network (RNet). The overall structure of the DCF model is shown in Fig. 2, where EbNet, subbands learning network (SLNet), GG, SA, and RNet denote the feature EbNet, the SLNet, the GG network, the SA network and the RNet, respectively.

In summary, the main contributions of this work are as follows:

1) A DCF model is proposed for MS and PAN images fusion. The model is deduced from the problem formulation of traditional MRA-based methods and is realized by exploring the intraspectral collaboration relationships and the spatial–spectral collaboration.
2) To avoid producing over-smooth fusion results, the model uses the wavelet decomposition to separate the subbands of PAN features and also explores the relationships among them. Furthermore, the model extracts the MS features band-by-band, and also utilizes the GCN to adjust the spectral relations, which is not mentioned in previous fusion works.
3) The spatial–spectral collaboration is realized in two different ways. First, the low-frequency subband of PAN image in the feature space are treated as guidance of the MS feature maps through a GG network. Then, the high-frequency subbands of PAN features are combined with the adjusted MS features to reconstruct the final MS image that has high resolution in both spatial and spectral domains.

The rest of this article is organized as follows: Section II introduces the motivation of the proposed method. Section III elaborates on the network structure and the fusion process. Experimental settings and corresponding results are shown in Section IV. Finally, the conclusions are drawn in Section V.

II. Motivations

Let $\mathbf{M} \in \mathbb{R}^{H/r\times W/r\times N_b}$ and $\mathbf{P} \in \mathbb{R}^{H\times W}$ represent the LR MS image that has N_b bands and the single-band HR PAN image, where r is the ratio of the resolution between MS and PAN. Traditional MRA-based fusion method can be formulated as

$$\begin{aligned}\widehat{\mathbf{M}}_b &= \widetilde{\mathbf{M}}_b + g_b(\mathbf{P} - \mathbf{P}_L)\\ &= \widetilde{\mathbf{M}}_b + g_b\mathbf{P}_H \end{aligned} \tag{1}$$

where $\widetilde{\mathbf{M}} \in \mathbb{R}^{H\times W\times N_b}$ and $\widehat{\mathbf{M}} \in \mathbb{R}^{H\times W\times N_b}$ represent the up-sampled MS image and the fusion result, and the subscript b in (1) indexes the bth band of them. $\mathbf{g} = [g_1, \ldots, g_{N_b}]$ is the injection coefficient. $\mathbf{P}_H$ and $\mathbf{P}_L$ denote the high-frequency and low-frequency components of the PAN image. Equation (1) indicates that the spectral information of the fused image comes from the up-sampled MS image and that the spatial details are from the high-frequency components of the PAN image.

Although a traditional MRA-based model has a straightforward interpretation of the fusion process, it has some limitations. First, Dong *et al.* has pointed out that the mapping from LR space to HR space has nonlinearity [32], however, (1) is essentially a linear model and cannot model the nonlinearity, therefore, this article generalizes the fusion process to the nonlinear case and conducts the fusion process in the feature space. Second, the spectral information of the fused image is consistent with that in the original MS image, however, it has some deviations from the up-sampled MS image, because available up-sampling methods are always based on some specific interpolation algorithms, and the inconsistent spectral information is inevitably introduced during the up-sampling process, which results in the spectral distortions in the fusion products. Fig. 3 shows the up-sampled MS image, the reference image and the spectral angle mapping (SAM) [40] between them. From the SAM map shown in Fig. 3(c), we can observe that there are large spectral variations between the up-sampled MS image and the reference image especially in the edges and texture areas. Hence, the spectral information should be adjusted before obtaining the fusion results. Due to the fact that the spectral curves of different objects are different, their spectral correlations among different spectral bands are different. In order to adjust the spectral relationships discriminatively, we should first map the images to the feature space. However, the 2-D convolutional operation extracts the features through aggregating the spatial-wise information and the channel-wise information, as shown in Fig. 4, and during this process, the information of bands is blended to each other, which makes the SA difficult. To overcome the difficulty, we extracted the feature of MS image band-by-band through a shared feature EbNet to preserve the information of each band, and then utilized a GCN to adjust the responses of channels in the feature space. Finally, we can

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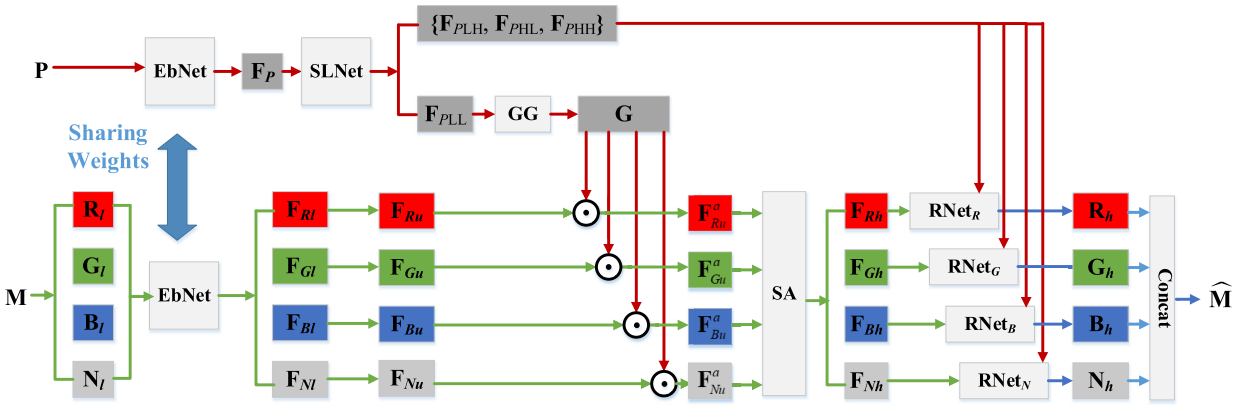


Fig. 2. Architecture of DCF model. EbNet, SLNet, GG network, SA network, and RNet$_x$: RNet of band x.

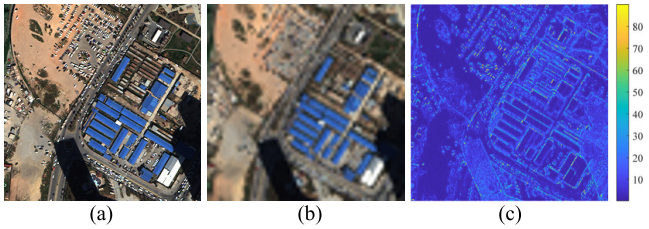


Fig. 3. Spectral relations of reference image and up-sampled MS image. (a) Reference image. (b) Up-sampled MS image. (c) SAM between reference image and up-sampled MS image.

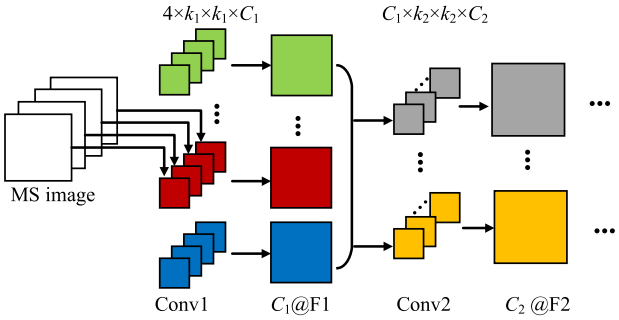


Fig. 4. 2-D convolutional operations, where C_1 at F1 means that the output of Conv1 has C_1 channels.

also observe from (1) that the high-frequency components of PAN image should be injected to preserve the spatial details, whereas available CNN-based methods pay little attention to the details and textures of the fusion results. To avoid producing over-smooth results, this article makes full use of the detail extraction ability of wavelet transform, and preserves the spatial details in the feature space.

Based on the above analyses, we combine the GCN with CNN to model the spatial–spectral collaboration relationship to finally develop a DCF model.

III. Proposed DCF Model

In this section, we provide the methodology of the proposed method. The overall structure of the model and the detailed operations of each subnetwork are presented as follows.

A. Overall Structure of DCF Model

As is shown in Fig. 2, the DCF model is composed of two branches of networks. The red arrows and the green arrows in Fig. 2 represent the operations in the PAN branch and MS branch, respectively, and the blue arrows denote the reconstruction operations. For the PAN branch, the features $\mathbf{F}_P$ are extracted by the feature EbNet. Then the four subbands are separated by the SWT. The low-frequency components $\mathbf{F}_{PLL}$ pass through a GG network to obtain the guidance $\mathbf{G}$, which is a crucial ingredient for the SA of MS image, while the high-frequency subbands $\{\mathbf{F}_{PLH}, \mathbf{F}_{PHL}, \mathbf{F}_{PHH}\}$ are treated as part of the inputs of the RNet. For the correspondence of features, the shared EbNet is utilized to map each band of MS image to the feature space. After extracting the features of each MS band, we up-sample them to obtain the MS features with the same size as the PAN features by the bilinear interpolation algorithm. The up-sampled MS features $\{\mathbf{F}_{Ru}, \mathbf{F}_{Gu}, \mathbf{F}_{Bu}, \mathbf{F}_{Nu}\}$ are guided by the guidance $\mathbf{G}$ to accomplish the SA operation. Finally, the adjusted MS features $\{\mathbf{F}_{Rh}, \mathbf{F}_{Gh}, \mathbf{F}_{Bh}, \mathbf{F}_{Nh}\}$ are combined with the high-frequency subbands of PAN features to act as the inputs of the RNet to achieve the mapping from HR feature space to HR image space. The detailed structure of each subnetwork is described as follows.

B. Feature EbNet

The preservation of information is very important in the fusion task, especially for remote sensing image fusion. Because the remote sensing images always have relatively lower spatial resolution and the type of objects are more diverse than natural images, the loss of information will result in more fusion errors. Therefore, the DenseNet structure [41] which has feature-reuse characteristic is utilized in the DCF model to construct the feature EbNet. Fig. 5 shows the detailed structure of EbNet, where DB represents the dense block.

EbNet is composed of an input convolutional layer (Conv1) and several DBs. Conv1 in Fig. 5 aims at extracting the shallow features and increasing the number of input channels of DB. The outputs of Conv1 are treated as the inputs of the first DB to further extract the deeper features, and they are also added

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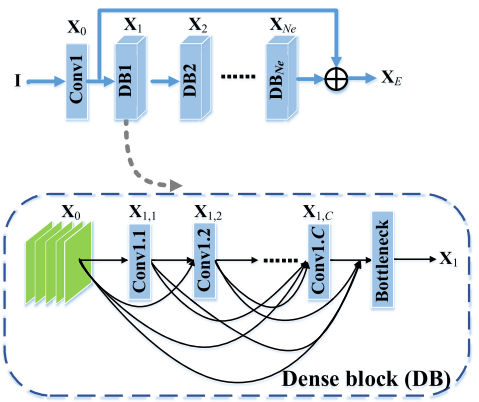


Fig. 5. Architecture of EbNet.

to the output features of the last DB by a skip connection. Assume that there are Ne DBs in EbNet, and in each DB, there are C convolutional layers as Fig. 5 shows. The operations in EbNet are formulated as follows:

$$\begin{aligned}\mathbf{X}_0 &= \sigma(\mathbf{W}_0 * \mathbf{I} + \mathbf{b}_0)\\ \mathbf{X}_d &= \mathbf{W}_d^B * \left[\mathbf{X}_{d-1}, \mathbf{X}_{d,1}, \mathbf{X}_{d,2}, \ldots, \mathbf{X}_{d,C}\right]\\ &\quad + \mathbf{b}_d^B, d = 1, \ldots, \mathrm{Ne}\\ \mathbf{X}_E &= \mathbf{X}_0 + \mathbf{X}_{\mathrm{Ne}}\end{aligned} \tag{2}$$

where $\mathbf{I} \in \mathbb{R}^{h\times w}$, $\mathbf{X}_0 \in \mathbb{R}^{h\times w\times m}$, $\mathbf{W}_0$ and $\mathbf{b}_0$ are the input–output, weights, and biases of the input convolutional layer Conv1. $\sigma(\cdot)$ is the nonlinear activation function. $[\mathbf{A}, \mathbf{B}]$ represents the concatenation of $\mathbf{A}$ and $\mathbf{B}$ in the channel dimension. $\mathbf{X}_{d,c} \in \mathbb{R}^{h\times w\times k}$ is the output of the cth convolutional layer in the dth DB, and $\mathbf{X}_d \in \mathbb{R}^{h\times w\times m}$ is the output of dth DB. $\mathbf{W}_d^B$ and $\mathbf{b}_d^B$ denote the weights and biases of the bottleneck layer in dth DB. $\mathbf{X}_E$ is the final output of EbNet. In dth DB, we have

$$\begin{aligned}\mathbf{X}_{d,c} &= \sigma\left(\mathbf{W}_{d,c} * \left[\mathbf{X}_{d-1}, \mathbf{X}_{d,1}, \ldots, \mathbf{X}_{d,c-1}\right] + \mathbf{b}_{d,c}\right),\\ &\quad c = 1, \ldots, C\end{aligned} \tag{3}$$

in which $\mathbf{W}_{d,c}$ and $\mathbf{b}_{d,c}$ are the weights and biases of the cth convolutional layer in the dth DB. The number of output channels of Conv1 and each DB is m, while that of each layer within a DB is k, and k is also called the growth rate in DenseNet. Compared with the original DenseNet structure, EbNet in DCF model not only has a global residual connection, but also has a bottleneck layer at the end of each DB to take feature maps of all previous layers as inputs and reduce the number of channels.

Let $\mathbf{F}_P \in \mathbb{R}^{H\times W\times m}$ represent the embedding features of a PAN image $\mathbf{P} \in \mathbb{R}^{H\times W}$. For the MS image $\mathbf{M} \in \mathbb{R}^{H/r\times W/r\times N_b}$, EbNet takes each band of $\mathbf{M}$ to obtain the corresponding embedding features. Take a four-band MS image as an example, the inputs of EbNet are $\{\mathbf{R}_l, \mathbf{G}_l, \mathbf{B}_l, \mathbf{N}_l\}$, and its outputs are $\{\mathbf{F}_{\mathrm{RI}}, \mathbf{F}_{\mathrm{GI}}, \mathbf{F}_{\mathrm{BI}}, \mathbf{F}_{\mathrm{NI}}\}$, where the subscripts R, G, B, and N represent the red band, green band, blue band, and the near infrared band, and the letter "l" denotes the LR, i.e., the size of them are r times smaller than that of PAN features $\mathbf{F}_P$. Theoretically, we use the weight sharing in feature EbNet because DCF model extracts MS features in a band-by-band fashion, and this network only focuses on extracting the spatial

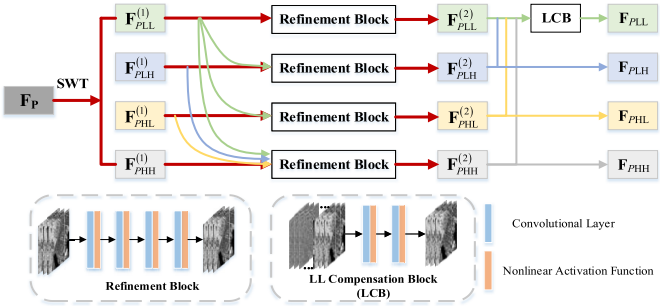


Fig. 6. Architecture of SLNet.

features. Furthermore, we also analyze the necessity of the weight sharing in subsequent ablation studies. In this article, except for the bottleneck layer that utilizes 1×1 kernels and does not have nonlinear activation function, the kernel size and nonlinear activation function of other convolutional layers are 3×3 and Leaky Rectified Linear Unit (LReLU) with $\alpha = 0.2$. The hyperparameters are empirically set as $m = 32$, $k = 16$, $\mathrm{Ne} = 4$, and $C = 5$, that is, EbNet is composed of four DBs, and each DB has five convolutional layers. The growth rate is 16 and the number of output channels of Conv1 and each bottleneck layer is 32.

C. Subbands Learning Network

The SLNet separates the subbands based on PAN features, and also conducts the subbands refinement operations on the four subbands. We separate the subbands of PAN in the feature domain for two reasons. First of all, owing to the specific structure of the EbNet, its output feature maps $\mathbf{F}_p$ encode the essential information in the original PAN image. And we conduct SWT on $\mathbf{F}_p$ to make the subband features corresponding to the MS features, because the reasonability of following GG network heavily depends on the correspondence relationship. Second, the separation in the feature domain makes the subbands connect with each other and also makes them learnable through the refinement block in the SLNet. The four sub-bands are represented by LL, HL, LH and HH, where the first character denotes the type of filters (L for low-pass filter and H for high-pass filter) along columns, and the second character denotes it along rows. The structure of SLNet is shown in Fig. 6, where the PAN features $\mathbf{F}_p$ are first decomposed into four subbands $\left\{\mathbf{F}_{PLL}^{(1)}, \mathbf{F}_{PLH}^{(1)}, \mathbf{F}_{PHL}^{(1)}, \mathbf{F}_{PHH}^{(1)}\right\}$ by SWT with the Haar wavelet, then the LL subband features $\mathbf{F}_{PLL}^{(1)}$ are used as part of the inputs for the HL, LH, and HH subbands refinement blocks. Fig. 7 shows the feature maps of the SLNet that are averaged along the channel dimension. From Fig. 7(a)–(d), we can observe that $\mathbf{F}_{PLL}^{(1)}$ can be used to refine $\left\{\mathbf{F}_{PLH}^{(1)}, \mathbf{F}_{PHL}^{(1)}, \mathbf{F}_{PHH}^{(1)}\right\}$, and $\mathbf{F}_{PLH}^{(1)}$ and $\mathbf{F}_{PHL}^{(1)}$ can also be utilized to refine $\mathbf{F}_{PHH}^{(1)}$, therefore, in the refinement block, we have

$$\begin{aligned}\mathbf{F}_{PLL}^{(2)} &= \mathcal{R}_{\mathrm{LL}}\left(\mathbf{F}_{PLL}^{(1)}\right)\\ \mathbf{F}_{PLH}^{(2)} &= \mathcal{R}_{\mathrm{LH}}\left(\left[\mathbf{F}_{PLL}^{(1)}, \mathbf{F}_{PLH}^{(1)}\right]\right)\\ \mathbf{F}_{PHL}^{(2)} &= \mathcal{R}_{\mathrm{HL}}\left(\left[\mathbf{F}_{PLL}^{(1)}, \mathbf{F}_{PHL}^{(1)}\right]\right)\\ \mathbf{F}_{PHH}^{(2)} &= \mathcal{R}_{\mathrm{HH}}\left(\left[\mathbf{F}_{PLL}^{(1)}, \mathbf{F}_{PLH}^{(1)}, \mathbf{F}_{PHL}^{(1)}, \mathbf{F}_{PHH}^{(1)}\right]\right)\end{aligned} \tag{4}$$

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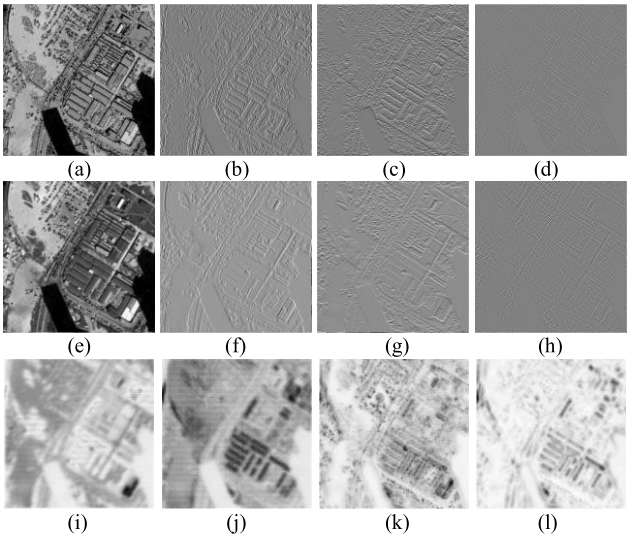


Fig. 7. Visualization of SLNet features and the up-sampled MS features that are averaged along the channel dimension, where (a)–(h) show the features of SLNet, and (i)–(l) show the up-sampled MS features. (a) $\mathbf{F}_{PLL}^{(1)}$. (b) $\mathbf{F}_{PLH}^{(1)}$. (c) $\mathbf{F}_{PHL}^{(1)}$. (d) $\mathbf{F}_{PHH}^{(1)}$. (e) $\mathbf{F}_{PLL}$. (f) $\mathbf{F}_{PLH}$. (g) $\mathbf{F}_{PHL}$. (h) $\mathbf{F}_{PHH}$. (i) $\mathbf{F}_{\mathrm{Ru}}$. (j) $\mathbf{F}_{\mathrm{Gu}}$. (k) $\mathbf{F}_{\mathrm{Bu}}$. (l) $\mathbf{F}_{\mathrm{Nu}}$.

where $\mathcal{R}_x(\cdot)$ denotes the refinement block of subband x. After the subbands refinement block, $\{\mathbf{F}_{PLH}^{(2)}, \mathbf{F}_{PHL}^{(2)}, \mathbf{F}_{PHH}^{(2)}\}$ are directly treated as the outputs of the SLNet, while $\mathbf{F}_{PLL}^{(2)}$ should be compensated by the LL compensation block (LCB) for they are considered as the inputs of GG network to generate the guidance maps, then we have

$$\begin{aligned}\mathbf{F}_{PLL} &= \mathrm{LCB}\left(\left[\mathbf{F}_{PLL}^{(2)}, \mathbf{F}_{PLH}^{(2)}, \mathbf{F}_{PHL}^{(2)}, \mathbf{F}_{PHH}^{(2)}\right]\right)\\ \mathbf{F}_{PLH} &= \mathbf{F}_{PLH}^{(2)}\\ \mathbf{F}_{PHL} &= \mathbf{F}_{PHL}^{(2)}\\ \mathbf{F}_{PHH} &= \mathbf{F}_{PHH}^{(2)}\end{aligned} \tag{5}$$

where $\mathrm{LCB}(\cdot)$ denotes the operations of LCB.

D. GG Network

The MS band features $\{\mathbf{F}_{\mathrm{Rl}}, \mathbf{F}_{\mathrm{Gl}}, \mathbf{F}_{Bl}, \mathbf{F}_{\mathrm{Nl}}\}$ obtained through EbNet have smaller size than PAN features. To facilitate subsequent fusion operations, they are first up-sampled by the bilinear interpolation algorithm to obtain the up-sampled features $\{\mathbf{F}_{\mathrm{Ru}}, \mathbf{F}_{\mathrm{Gu}}, \mathbf{F}_{\mathrm{Bu}}, \mathbf{F}_{\mathrm{Nu}}\}$, where the letter "$u$" in the subscript means that they are up-sampled version of MS band features. After the LL subband compensation operation, the LL subband features of PAN image can be used for the GG network. It can be observed from Fig. 7(e) and Fig. 7(i)–(l) that the compensated LL components of PAN features can better describe the appearances and profiles of objects than MS features. Owing to the fact that different objects have different spectral characteristics, i.e., the spectral relationships among different spectral bands are different, these objects should be separated discriminatively before SA. The GG network used in DCF model is motivated by the attention mechanism [42]–[44] in the field of computer vision. From previous analyses, we know that the low-frequency components of PAN features can well describe the appearances and profiles of objects, then

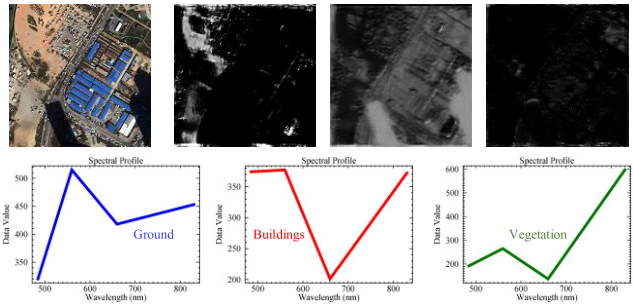


Fig. 8. Examples of guidance maps, where the first subfigure is the referenced MS image, and others are the features of the guidance which show high confidence of a specific object (ground, buildings, and vegetation from left to right).

GG network extracts the attention maps from low-frequency components of PAN features to construct the guidance which can be used to guide the subsequent SA of MS features.

The guidance can be obtained by

$$\mathbf{G} = \varphi(\mathbf{F}_{PLL}) \tag{6}$$

where $\varphi(\cdot)$ denotes the composition of GG operations. GG network takes the low-frequency components $\mathbf{F}_{PLL}$ as input, and obtains the guidance through a series of convolutional layers. It should be noted that the nonlinear activation function of the last convolutional layer is the sigmoid function. Fig. 8 shows the referenced MS image and several feature maps of $\mathbf{G}$. From Fig. 8, we can observe that each channel has a typical object that has higher spectral response values than other objects: the feature maps from left to right have higher response values for ground, buildings and vegetation, respectively, and the spectral profiles of them are totally different. Due to the fact that the GG process does not change the number of channels of feature maps, the up-sampled MS features are then combined with attention maps through

$$\begin{aligned}\mathbf{F}_{\mathrm{Ru}}^{a} &= \mathbf{G} \odot \mathbf{F}_{\mathrm{Ru}}\\ \mathbf{F}_{\mathrm{Gu}}^{a} &= \mathbf{G} \odot \mathbf{F}_{\mathrm{Gu}}\\ \mathbf{F}_{\mathrm{Bu}}^{a} &= \mathbf{G} \odot \mathbf{F}_{\mathrm{Bu}}\\ \mathbf{F}_{\mathrm{Nu}}^{a} &= \mathbf{G} \odot \mathbf{F}_{\mathrm{Nu}}\end{aligned} \tag{7}$$

where "$\odot$" is the element-wised product and the letter "a" in the superscript represents that they are the MS features mapped by the guidance map. The GG network in this article consists of three convolutional layers whose kernel size is 3×3 for the first two layers and 1×1 for the last layer. Moreover, the nonlinear function of the first two layers is LReLU with $\alpha = 0.2$ and that of the last layer is sigmoid function.

E. SA Network

The SA network is composed of several graph convolutional (GC) layers, and the inputs of each GC layer are the adjacent matrix and the features of previous layer. As mentioned above, the outputs of the GG network are guided MS features $\mathbf{F}_{\mathrm{Ru}}^{a} \in \mathbb{R}^{H\times W\times m}$, $\mathbf{F}_{\mathrm{Gu}}^{a} \in \mathbb{R}^{H\times W\times m}$, $\mathbf{F}_{\mathrm{Bu}}^{a} \in \mathbb{R}^{H\times W\times m}$ and $\mathbf{F}_{\mathrm{Nu}}^{a} \in \mathbb{R}^{H\times W\times m}$. First, we vectorize the ith channel of the guided MS features, and they are denoted as $\mathbf{y}_{R,i}^{0} \in \mathbb{R}^{\mathrm{HW}\times 1}$,

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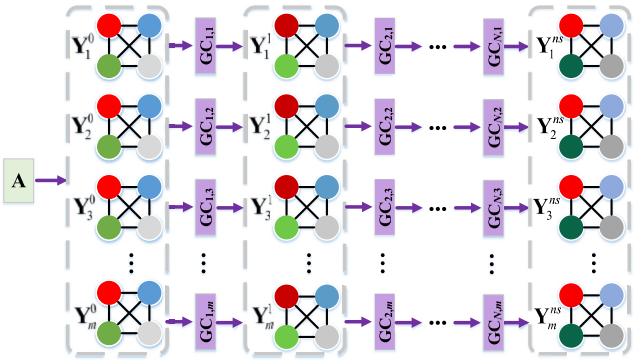


Fig. 9. Architecture of SA network, where $GC_{x,y}$ represents the yth graph of the xth GC layer.

$\mathbf{y}_{G,i}^0 \in \mathbb{R}^{\mathrm{HW}\times 1}$, $\mathbf{y}_{B,i}^0 \in \mathbb{R}^{\mathrm{HW}\times 1}$ and $\mathbf{y}_{N,i}^0 \in \mathbb{R}^{\mathrm{HW}\times 1}$ $(i = 1,\ldots,m)$. Then the ith graph input of the first GC layer is $\mathbf{Y}_i^0 = \left[\mathbf{y}_{R,i}^0, \mathbf{y}_{G,i}^0, \mathbf{y}_{B,i}^0, \mathbf{y}_{N,i}^0\right] \in \mathbb{R}^{\mathrm{HW}\times 4}$. It should be noted that due to the spectral correlations exist among the four bands of the MS image, the adjacent matrix $\mathbf{A}$ is defined as

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}. \tag{8}$$

There are m graphs in total, and they share the adjacent matrix $\mathbf{A}$ shown in (8), which means that they have the same graph structure. Therefore, the GC operation is formulated as

$$\mathbf{Y}_i^n = \sigma\left(\mathbf{A}\mathbf{Y}_i^{n-1}\mathbf{W}_i^n\right), \quad n = 1,\ldots,\mathrm{Ns},\ i = 1,\ldots,m \tag{9}$$

where $\mathbf{Y}_i^n$ and $\mathbf{W}_i^n$ are the output and weight of the nth GC layer for the ith graph. $\sigma(\cdot)$ is the nonlinear activation function. In this article, the SA network is composed of four GC layers and the nonlinear function is also LReLU with $\alpha = 0.2$. The detailed architecture of SA network is shown in Fig. 9.

F. Reconstruction Network

After the EbNet, SLNet, GG network, and the SA network, MS features are combined with the low-frequency components of PAN features, and these combined features are also adjusted through a series of GC layers to obtain the adjusted HR MS features $\{\mathbf{F}_{\mathrm{Rh}}, \mathbf{F}_{\mathrm{Gh}}, \mathbf{F}_{\mathrm{Bh}}, \mathbf{F}_{\mathrm{Nh}}\}$. However, the spatial details contained in the high-frequency subbands of PAN features are not considered in the these adjusted MS features, therefore, the RNet takes the high-frequency components of PAN features and the adjusted MS features as inputs, and maps the fused features from the feature space to the image space to obtain each band image of the final fusion product. First of all, we use each of the adjusted MS features to replace the LL subband of PAN features, and conduct the inverse SWT on the subbands

$$\begin{aligned} \mathbf{F}_{\mathrm{Rh}}^{(1)} &= \mathrm{ISWT}(\mathbf{F}_{\mathrm{Rh}}, \mathbf{F}_{PLH}, \mathbf{F}_{PLH}, \mathbf{F}_{PLH}) \\ \mathbf{F}_{\mathrm{Gh}}^{(1)} &= \mathrm{ISWT}(\mathbf{F}_{\mathrm{Gh}}, \mathbf{F}_{PLH}, \mathbf{F}_{PLH}, \mathbf{F}_{PLH}) \\ \mathbf{F}_{\mathrm{Bh}}^{(1)} &= \mathrm{ISWT}(\mathbf{F}_{\mathrm{Bh}}, \mathbf{F}_{PLH}, \mathbf{F}_{PLH}, \mathbf{F}_{PLH}) \\ \mathbf{F}_{\mathrm{Nh}}^{(1)} &= \mathrm{ISWT}(\mathbf{F}_{\mathrm{Nh}}, \mathbf{F}_{PLH}, \mathbf{F}_{PLH}, \mathbf{F}_{PLH}) \end{aligned} \tag{10}$$

where $\mathrm{ISWT}(\cdot)$ represents the inverse SWT. Let $f_R(\cdot)$, $f_G(\cdot)$, $f_B(\cdot)$, and $f_N(\cdot)$ represent the mappings of RNet for each band, then we have

$$\begin{aligned} \mathbf{R}_h &= f_R\left(\mathbf{F}_{\mathrm{Rh}}^{(1)}\right) \\ \mathbf{G}_h &= f_G\left(\mathbf{F}_{\mathrm{Gh}}^{(1)}\right) \\ \mathbf{B}_h &= f_B\left(\mathbf{F}_{\mathrm{Bh}}^{(1)}\right) \\ \mathbf{N}_h &= f_N\left(\mathbf{F}_{\mathrm{Nh}}^{(1)}\right). \end{aligned} \tag{11}$$

After obtaining each band of HR MS images, we concatenate them along the spectral dimension to get the final fusion result $\widehat{\mathbf{M}}$

$$\widehat{\mathbf{M}} = [\mathbf{R}_h, \mathbf{G}_h, \mathbf{B}_h, \mathbf{N}_h]. \tag{12}$$

The RNets share the same structures, but they do not share their weights. Each RNet is composed of two convolutional layers, where the kernel size of the first layer is 3×3 and that of the second layer is 1×1. Moreover, the first convolutional layer also uses LReLU with $\alpha = 0.2$ as its nonlinear activation function.

G. Loss Function

Most of the available DL-based MS and PAN images fusion models utilized the mean squared error (MSE) as their loss function. However, Sajjadi *et al.* [31] pointed out that although MSE loss could obtain a higher peak-signal-to-noise (PSNR) value, the visual inspection of the reconstruction result was over-smooth. Furthermore, Scarpa *et al.* [25] conducted several comparison experiments and drew the conclusion that their model who used the mean absolute error (MAE) loss had a better fusion result than that used the MSE loss, and the whole model also converged faster. Inspired by their works, we also take the MAE loss as the loss function in the DCF model, that is,

$$\mathcal{L}\left(\widehat{\mathbf{M}}, \mathbf{R}\right) = \left\|\widehat{\mathbf{M}} - \mathbf{R}\right\|_1 \tag{13}$$

where $\mathbf{R} \in \mathbb{R}^{H\times W\times N_b}$ denotes the reference image.

IV. Experimental Results and Discussions

In order to verify the effectiveness of the proposed method, DCF model is compared with seven state-of-the-art MS and PAN image fusion methods including four traditional methods and three DL-based methods. The traditional methods used in this section are Gram–Schmidt Adaptive (GSA) method [12], Modulation Transfer Function-based Generalized Laplacian Pyramid method with High-Pass Modulation (MTF-GLP-HPM) method [45], Patch Normalization-based Two-Step Sparse Coding (pn-TSSC) method [46] and Variational Pan-Sharpening with Local Gradient Constraints (VP-LGC) method [21], where GSA [12] belongs to the CS-based fusion methods, MTF-GLP-HPM is an MRA-based fusion method, pn-TSSC [46] and VP-LGC [21] methods are MBO-based fusion methods. The compared DL-based methods are Deep Metric Learning and Geometric Multimanifold Embedding (DML-GMME) method [23], Target-Adaptive Pan-sharpening

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Neural Network (TA-PNN) [25], and Pan-sharpening Network (PanNet) [28]. Because PanNet [28] is trained on the eight-band MS images, in the experiments of the four-band images, we only choose the network structure corresponding to the Blue, Green, Red, and Near infrared bands, and retrained the network on our training set.

A. Data Sets and Training Details

Because the MS images are not only four-band images, we train our model based on the four-band data set and eight-band dataset,[1] respectively. The four-band data set is acquired by WorldView-4 (WV-4) satellite, and the eight-band data set is obtained from the WorldView-2 (WV-2) satellite. The spatial resolution of PAN images is 0.3 m for WV-4 data set and 0.5 m for WV-2 data set, and the spatial resolution of MS images is four times lower than that of PAN images. That is, we have $r = 4$. For each data set, there are totally 48 900 patch pairs randomly cropped from the original large-scale data, where 39120 patch pairs are used for training and the rest 9780 patches for validation. The size of MS patch is $32 \times 32 \times 4$ ($32 \times 32 \times 8$ for WV-2 data set). Afterward, we crop another 200 test image pairs that are not included in the training set and the validation set to form a test set. The test MS images have the size of $256 \times 256 \times 4$ ($256 \times 256 \times 8$ for WV-2 data set) and PAN images are composed of 1024×1024 pixels. Experiments are conducted at the reduced resolution and the full resolution. In the reduced resolution case, according to Wald's protocol [47], all the original images are filtered by a 7×7 Gaussian smoothing kernel and are down-sampled by a factor of 4, then the down-sampled MS and PAN images constitute the inputs of the fusion methods and the original MS images are treated as the reference images. The index evaluations, four-band extension of universal image quality index (Q4), eight-band extension of universal image quality index (Q8) [48], SAM [40] and relative dimensionless global error in synthesis [Erreur Relative Globale Adimensionnelle de Synthèse (ERGAS)] [49] are used to evaluate the fusion results. In the full resolution case, the original MS and PAN images are directly considered as the inputs of the algorithms, and the spectral distortion index (D_λ), the spatial distortion index (D_s) and the jointly spectral and spatial quality index [Quality with No Reference (QNR)] indices are used for evaluations [50]. We call the proposed model trained on four-band MS images DCF-4, and that trained on eight-band MS images DCF-8. Owing to that the characteristics of remote sensing images from different satellites are different, we also test the trained DCF-4 model on GeoEye-1 and QuickBird data sets to show the generalization ability of the model. The test GeoEye-1 and QuickBird data sets are also composed of $256 \times 256 \times 4$ MS images and 1024×1024 PAN images. The spatial resolution is 0.5 m for GeoEye-1 PAN images and 0.6 m for QuickBird PAN images.

In order to train the model, reference images are required, therefore, we filter the original MS and PAN images in the training set according to Wald's protocol mentioned before, and the original MS images are treated as the reference images. During training, the learning rate is initialized to 10^{-3}, and it is divided by ten every 100 epochs. We use the Adam optimizer to train the model, where $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The batch size is 32 and all the weights are initialized with the Xavier initialization [51]. Implementation is supported by the PyTorch [52] framework, and the training is conducted on an Nvidia GTX 1080Ti GPU. After 300 epochs, the training process is terminated. It took roughly 12 h to train the DCF-4 model and 18 h to train the DCF-8 model. The number of float-point operations (FLOPs) and the total number of parameters of our model are about 2.33 GFLOPs and 1.36×10^5, respectively.

[1] Available online: https://digitalglobe.com/samples?search=Imagery

TABLE I
PARAMETER ANALYSIS ON WORDVIEW-4 DATA SET

N_e @ m @ N_s	Q4	SAM	ERGAS
4 @ 32 @ 4	0.945 ± 0.016	2.377 ± 0.752	2.491 ± 0.855
5 @ 32 @ 4	0.940 ± 0.018	2.746 ± 0.803	2.482 ± 0.810
3 @ 32 @ 4	0.912 ± 0.019	3.003 ± 0.948	3.215 ± 0.911
4 @ 16 @ 4	0.886 ± 0.020	4.810 ± 0.874	4.952 ± 1.054
4 @ 64 @ 4	0.951 ± 0.018	2.255 ± 0.838	2.361 ± 0.902
4 @ 32 @ 3	0.897 ± 0.021	4.234 ± 1.307	4.776 ± 0.831
4 @ 32 @ 5	0.947 ± 0.019	2.399 ± 0.973	2.442 ± 0.965

B. Parameter Analysis

In order to investigate the effects of model parameters on the fusion results, some experiments were conducted to analyze the model parameters of DCF-4, including N_e, C, m, and N_s, which represent the number of DBs in EbNet, the number of convolutional layers in each DB, the number of output channels of EbNet, and the number of GC layers in SA network. Due to the fact that both N_e and C relate to the depth of the network and their effects on the model are similar, we only discuss the case where N_e is increased or decreased with a fixed value of C. Table I shows the fusion results of DCF model on different configurations.

First, we changed the value of N_e with other parameters fixed to investigate the impact of EbNet's depth on the fusion results. The comparisons are shown in the first three rows in Table I. As is mentioned above, EbNet plays the role of mapping the images from image space to the embedding feature space, and theoretically, the deeper the EbNet is, the stronger its feature extraction ability is. But only ERGAS obtains a better value when $N_e = 5$, which may result from the unstable gradient or the overfitting during the training process for a deeper network structure. Therefore, we choose $N_e = 4$.

The first row, the fourth row, and the fifth row in Table I demonstrate the effect of m on the fusion results. It can be observed that the best values of all evaluation indices are obtained when $m = 64$. However, m not only affects the output of EbNet, but also relates to the number of graphs in the SA network, and a larger m means more graphs in SA network, which results in a larger scale of parameters to be trained. At the same time, compared with $m = 16$, the fusion results get a great promotion when $m = 32$. Therefore, in order to obtain a satisfactory fusion result with as fewer parameters as possible, m is set to 32.

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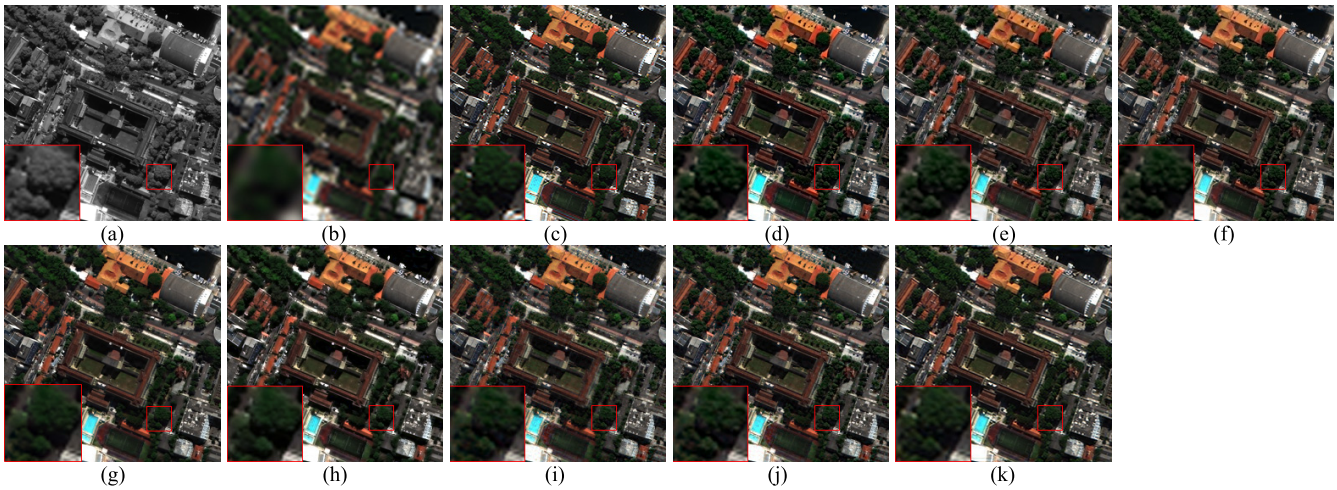


Fig. 10. Visual comparison among different fusion methods on the four-band WV-4 data set at the reduced resolution (256 × 256). (a) PAN. (b) EXP. (c) Reference. (d) GSA [12]. (e) MTF-GLP-HPM [45]. (f) pn-TSSC [46]. (g) VP-LGC [21]. (h) DML-GMME [23]. (i) TA-PNN [25]. (j) PanNet [28]. (k) DCF-4.

Finally, we investigate the effect of N_s. The first row, the sixth row, and the seventh row in Table I show the results of comparison experiments. From Table I, we know that the fusion model cannot produce a well fusion result when $N_s = 3$, whereas, the improvements are significant from $N_s = 3$ to $N_s = 4$, which motivates us to stack more GC layers. However, when $N_s = 5$, the improvements of evaluation indices are no longer evident, then we draw the conclusion that it is not necessary to stack more GC layers in DCF model, because they may bring more parameters to be trained and less improvement on evaluation indices. The number of GC layers is set to 4.

TABLE II
EVALUATION INDEXES OF DIFFERENT FUSION METHODS ON WV-4 DATA AT THE REDUCED RESOLUTION

Methods	Q4	SAM	ERGAS
Ideal	1	0	0
GSA [12]	0.8946	4.4028	3.8241
MTF-GLP-HPM [45]	0.8813	4.2424	4.0230
PN-TSSC [46]	0.9018	4.1104	3.5003
VP-LGC [21]	0.9216	4.6104	3.9595
DML-GMME [23]	0.9354	3.1227	2.6781
TA-PNN [25]	0.9219	3.9220	2.7567
PanNet [28]	0.9361	3.3179	2.8515
DCF-4	**0.9607**	**2.4897**	**2.2403**

C. Experimental Results at Reduced Resolution

In this section, the DCF-4 (DCF-8) model is compared with several fusion methods at the reduced resolution.

1) DCF-4 Performance on the Four-Band Images: Fig. 10 shows the fusion results of different fusion methods, where EXP is the interpolated MS image that is interpolated by a polynomial kernel with 23 coefficients [10]. In order to show the differences evidently, we zoomed in certain areas of the fusion results, and from the amplified area in Fig. 10(c), we can observe that the cars beside the trees are white, however, in Fig. 10(g), (i), and (j), these cars seem to be a little dim and the color of them tends to be blended with that of the roads, which means that the fusion results of VP-LGC [21], TA-PNN [25], and PanNet [28] methods suffer from serious spectral distortion. Meanwhile, the results shown in Figs. 10(d)–(f) have some artifacts in the amplified areas where the trees have some white edges. Although the DML-GMME [23] method obtains comparable fusion products, it can also be observed from the trees in the amplified areas that the blurry effect appears in its result. Therefore, our proposed method outperforms other compared methods in terms of spectral preservation and spatial enhancement. The quantitive evaluation indexes were also computed and demonstrated in Table II, where the best values are shown in bold. From Table II, we can observe that the proposed method obtains the best values in Q4, SAM, and ERGAS indexes. For Q4 index, PanNet [28] obtains the second best, while for SAM and ERGAS indexes, DML-GMME [23] method obtains the second best, which are in consistent with our visual analysis that DML-GMME [23] method has stronger spectral preservation ability but weaker spatial enhancement ability, and the proposed method performs the best in this data set.

2) DCF-8 Performance on the Eight-Band Images: Experimental results and quality evaluations on the eight-band WV-2 data set at the reduced resolution are shown in Fig. 11 and Table III. It can be observed from Figs. 11(d)–(e) and Fig. 11(j) that the fusion results of GSA [12], MTF-GLP-HPM [45], and PanNet [28] suffer from serious spectral distortions, where the grounds look like green. At the same time, the result of VP-LGC [21] method has some weird colors especially the cars in the amplified area. The spectral preservation abilities of pn-TSSC [46] and DML-GMME [23] are better than other methods, however, there are some artifacts in the result of DML-GMME [23] method. From Fig. 11(i), the result of TA-PNN [25] is blurry. Though both pn-TSSC [46] method and DCF-8 model have better fusion results, the proposed method performs much better in terms of the spatial

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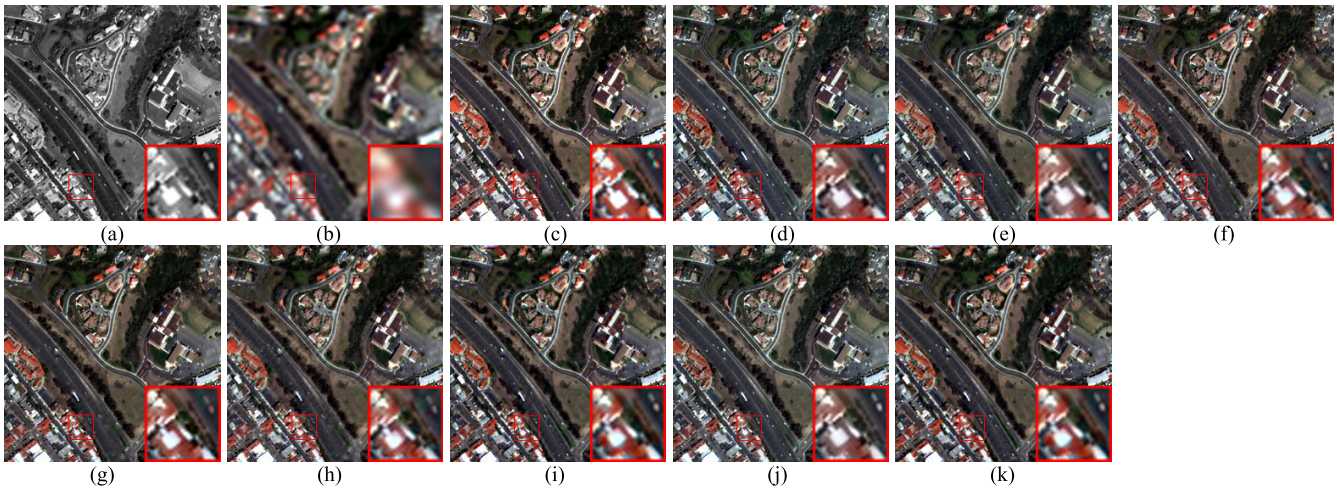


Fig. 11. Visual comparison among different fusion methods on the eight-band WV-2 data set at the reduced resolution (256 × 256). (a) PAN. (b) EXP. (c) Reference. (d) GSA [12]. (e) MTF-GLP-HPM [45]. (f) pn-TSSC [46]. (g) VP-LGC [21]. (h) DML-GMME [23]. (i) TA-PNN [25]. (j) PanNet [28]. (k) DCF-8.

TABLE III
EVALUATION INDEXES OF DIFFERENT FUSION METHODS ON WV-2 DATA AT THE REDUCED RESOLUTION

Methods	Q8	SAM	ERGAS
Ideal	1	0	0
GSA [12]	0.8816	5.6136	3.6942
MTF-GLP-HPM [45]	0.8920	4.8690	3.2127
PN-TSSC [46]	0.9019	5.2371	3.2729
VP-LGC [21]	0.9172	4.3885	2.9542
DML-GMME [23]	0.8934	5.1847	3.2500
TA-PNN [25]	0.8832	5.4941	3.8860
PanNet [28]	0.8888	4.9136	3.3308
DCF-8	**0.9274**	**3.9527**	**2.8871**

TABLE IV
EVALUATION INDEXES OF DIFFERENT FUSION METHODS ON WV-4 DATA AT THE FULL RESOLUTION

Methods	D_λ	D_s	QNR
Ideal	0	0	1
GSA [12]	0.0531	0.0650	0.8853
MTF-GLP-HPM [45]	0.0704	0.1071	0.8300
PN-TSSC [46]	0.0252	0.0406	0.9352
VP-LGC [21]	0.0090	0.0513	0.9402
DML-GMME [23]	0.0144	0.0385	0.9477
TA-PNN [25]	0.0468	0.0487	0.9068
PanNet [28]	0.0198	0.0448	0.9363
DCF-4	**0.0079**	**0.0294**	**0.9629**

enhancement and the spectral preservation abilities, which is in consistent with the index evaluations shown in Table III.

D. Experimental Results at Full Resolution

Apart from the experiments at the reduced resolution, we also conducted experiments at the full resolution, where no down-sampling operations were performed on the source images. The sizes of the images to be fused were 1024 × 1024 for PAN and 256 × 256 × 4 (256 × 256 × 8 for WV-2 data set) for MS. In the full-resolution experiment, there are no reference images, then the quality indexes D_λ, D_s, and QNR that do not need reference images are used for quantitive evaluations.

1) DCF-4 Performance on the Four-Band Images: Fig. 12 shows the fusion results of different methods at the full resolution. It can be found from Fig. 12 that results of MTF-GLP-HPM [45] and VP-LGC [21] are seriously blurry. The quality index D_s in Table IV also verifies the conclusion. The spectral distortion appears in the results of GSA [12], pn-TSSC [46], DML-GMME [23] and TA-PNN [25], where the spectral artifacts also exist in the results of pn-TSSC [46] and TA-PNN [25]. There are some spatial artifacts appear in the PanNet [28] fusion result, especially in the areas of the white roofs. It is obvious that our proposed DCF-4 outperforms other comparison methods both in the spatial and spectral domains. Table IV provides the quantitative evaluations of all comparison methods together with the proposed one.

TABLE V
EVALUATION INDEXES OF DIFFERENT FUSION METHODS ON WV-2 DATA AT THE FULL RESOLUTION

Methods	D_λ	D_s	QNR
Ideal	0	0	1
GSA [12]	0.0584	0.0563	0.8886
MTF-GLP-HPM [45]	0.0680	0.0652	0.8712
PN-TSSC [46]	0.0430	0.0396	0.9191
VP-LGC [21]	0.0394	0.0327	0.9292
DML-GMME [23]	0.0283	0.0313	0.9414
TA-PNN [25]	0.0352	0.0328	0.9332
PanNet [28]	0.0394	0.0240	0.9375
DCF-8	**0.0173**	**0.0119**	**0.9711**

2) DCF-8 Performance on the Eight-Band Images: The fusion results and index evaluations on the eight-band WV-2 data are shown in Fig. 13 and Table V. In the aspect of visual inspection, the DML-GMME [23] result seems to be a little blurry, while the fusion results of GSA [12], MTF-GLP-HPM [45] and TA-PNN [25] are over-enhanced, which reflect in the spaces between the two cars in the roads.

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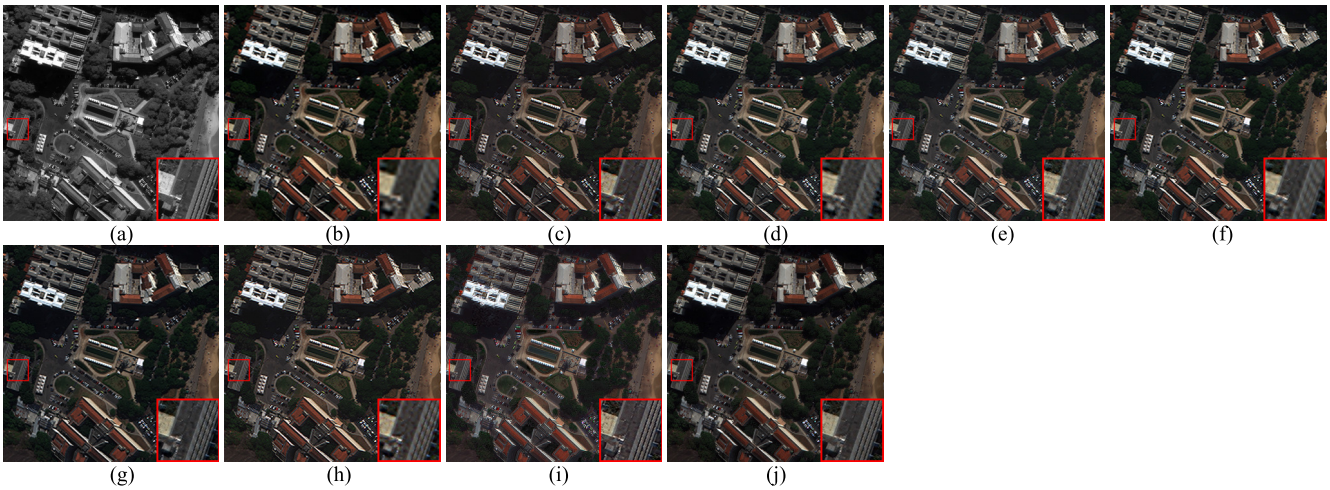


Fig. 12. Visual comparison among different fusion methods on the four-band WV-4 data set at the full resolution (1024 × 1024). (a) PAN. (b) EXP. (c) GSA [12]. (d) MTF-GLP-HPM [45]. (e) pn-TSSC [46]. (f) VP-LGC [21]. (g) DML-GMME [23]. (h) TA-PNN [25]. (i) PanNet [28]. (j) DCF-4.

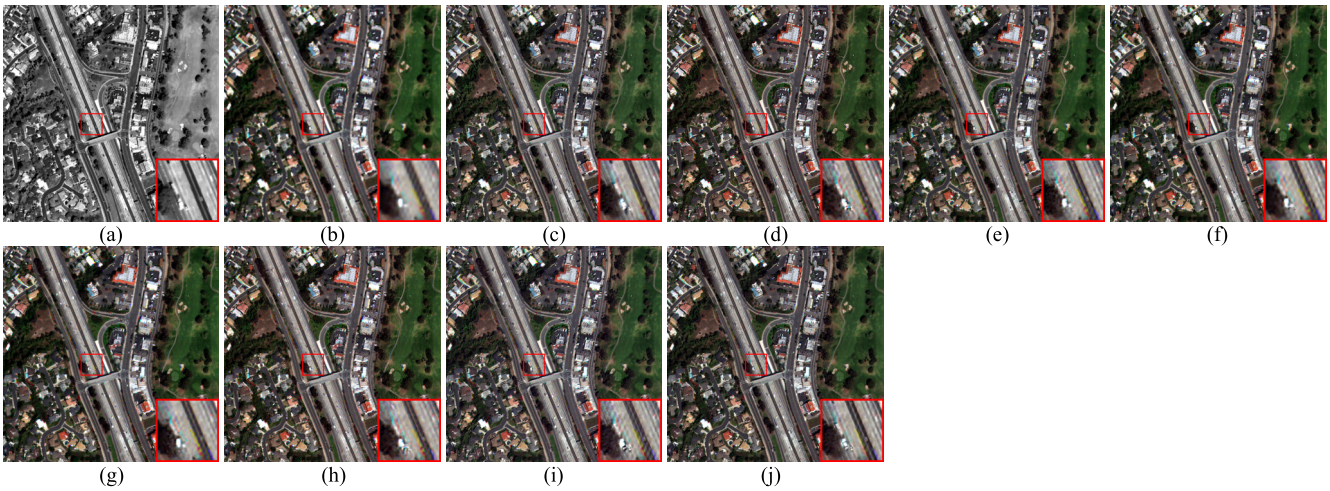


Fig. 13. Visual comparison among different fusion methods on the eight-band WV-2 data set at the full resolution (1024 × 1024). (a) PAN. (b) EXP. (c) GSA [12]. (d) MTF-GLP-HPM [45]. (e) pn-TSSC [46]. (f) VP-LGC [21]. (g) DML-GMME [23]. (h) TA-PNN [25]. (i) PanNet [28]. (j) DCF-8.

The pn-TSSC [46] and VP-LGC [21] results suffer from serious spectral distortion, and there are weird colors in the amplified areas. Moreover, the aliasing effects appear in the result of PanNet [28]. In the aspect of quantitive evaluations, four DL-based fusion methods obtain better index values compared with traditional fusion methods as demonstrated in Table V, which verifies the effectiveness of deep neural networks on the remote sensing image fusion task. From the visual inspection and the quality indexes, we can draw the conclusion that our proposed method has both high spectral-preservation and spatial-enhancement abilities, and can make a balance between spectral and spatial resolutions.

E. Ablation Studies

To evaluate the effects of different components on the final fusion results, in this section, we conduct a comprehensive ablation study, and in order to eliminate the impact of other factors, all configurations are kept in the same. We take the network structure shown in Fig. 2 as our *Baseline* structure. In general, the complete ablation studies should contain the following four cases:

Case 1 *(Different EbNet Weights)*: The MS and PAN branches did not share the weights of EbNet in this case.

Case 2 *(Without SLNet)*: The high-frequency subbands features $\{\mathbf{F}^{(1)}_{P\mathrm{LH}}, \mathbf{F}^{(1)}_{P\mathrm{HL}}, \mathbf{F}^{(1)}_{P\mathrm{HH}}\}$ were directly taken as the inputs of RNets and low-frequency subband $\mathbf{F}^{(1)}_{P\mathrm{LL}}$ was taken as the input of GG network without any refinement.

Case 3 *(Without SA Network)*: In this case, we abandoned the SA network, and directly took $\left\{\mathbf{F}^{a}_{\mathrm{Ru}}, \mathbf{F}^{a}_{\mathrm{Gu}}, \mathbf{F}^{a}_{\mathrm{Bu}}, \mathbf{F}^{a}_{\mathrm{Nu}}\right\}$ as the inputs of RNets.

Case 4 *(Without GG Network)*: We had discarded the GG network, and directly utilized $\mathbf{F}_{P\mathrm{LL}}$ to guide the SA network. However, it was found that in this configuration, we could not obtain effective fusion results, and the results were truly cluttered and discontinuous, therefore, in the ablation studies, we neglected the first two layers of GG network, and only retained the last convolutional layer, i.e., the 1 × 1

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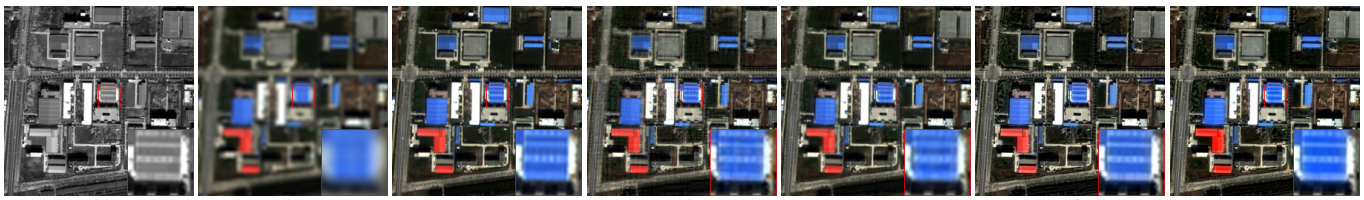


Fig. 14. Visual comparison among DL-based fusion methods on QuickBird data set at the reduced resolution (256 × 256). (a) PAN. (b) EXP. (c) Reference. (d) DML-GMME [23]. (e) TA-PNN [25]. (f) PanNet [28]. (g) DCF-4.

TABLE VI
ABLATION STUDIES ON WORDVIEW-4 DATA SET

Model	Q4	SAM	ERGAS
Baseline	0.945 ± 0.016	2.377 ± 0.752	2.491 ± 0.855
Case 1	0.834 ± 0.038	6.746 ± 1.724	7.001 ± 0.982
Case 2	0.857 ± 0.042	5.958 ± 0.830	6.584 ± 0.933
Case 3	0.902 ± 0.035	5.975 ± 1.087	5.940 ± 0.984
Case 4	0.895 ± 0.028	5.734 ± 1.579	5.449 ± 0.791

convolutional layer with the sigmoid function to guide the SA network.

Table VI summarizes the results of ablation studies. By comparing *Baseline* and *Case* 1 in Table VI, we find that the weight sharing of EbNet is necessary for it can guarantee the correspondence of PAN and MS band features in the feature space. Because DCF model extracts MS features in a band-by-band fashion. The comparison between *Baseline* and *Case* 2 validates the necessity of SLNet. The SLNet refines the $\{\mathbf{F}_{PLH}, \mathbf{F}_{PHL}, \mathbf{F}_{PHH}\}$ subbands to make them more accurate, and also compensates the $\mathbf{F}_{PLL}$ subband to adapt for the guidance of MS features. Furthermore, the SA network is also important for it influences the spectral resolution, which can be seen from the comparison between the *Baseline* and *Case 3*, especially on the SAM index. From the description of *Case 4*, we know that the product of GG network, i.e., the guidance, acts as a weight map of the up-sampled MS features because it can amplify the features with high response values and also suppress others to facilitate the SA. If we absolutely abandon the GG network, $\mathbf{F}_{PLL}$ cannot serve on the weight maps, which results in the clustered results. Consequently, the sigmoid nonlinear activation in this part is necessary. Therefore, the network configuration in *Case* 4 keeps the 1×1 convolutional layer with the sigmoid function. We can observe from the comparison between *Baseline* and *Case* 4 in Table VI that the GG network brings an averaged 0.05 promotion in terms of Q4 index, and also improves the SAM and ERGAS indexes by 3.0.

These studies show that each module in our DCF model brings individual improvements, and all of them work together to produce the excellent fusion results.

F. Runtime Comparisons

In this section, we conduct the runtime comparison experiments on the images with the spatial size of 256 × 256. The test experiments are conducted on the laptop with Intel Core i7 2.60-GHz processor with 16-GB RAM. Table VII shows the runtime of all the fusion methods. It is obvious that all the fusion methods have tolerable runtimes, and traditional CS- or MRA-based methods are the most time-efficient, while the MBO-based methods need more times. In the DL-based fusion methods, DML-GMME [23] requires more time because the reconstruction process of it is similar with traditional MBO-based methods. Among other DL-based fusion models, TA-PNN [25] is faster than PanNet [28] and DCF for its simplest architecture. The time consumptions of all the fusion methods are in a tolerated range.

G. Generalization to Other Satellite Data Sets

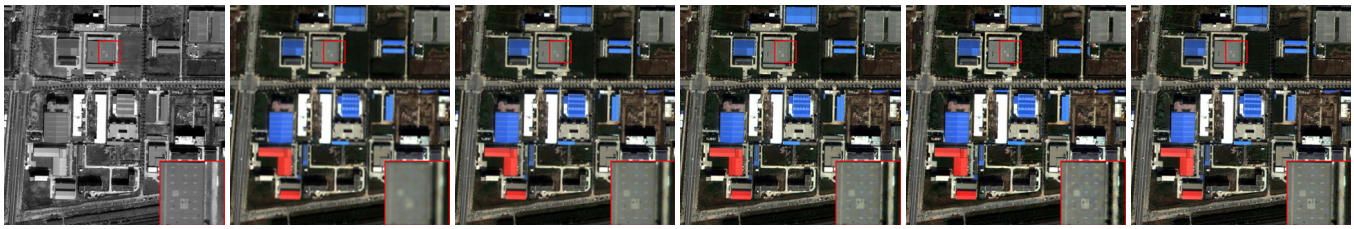
In this section, we use the DCF-4 model trained on the WV-4 data set to fuse the images obtained by other satellites to validate the generalization ability of the proposed method. Furthermore, DML-GMME [23], TA-PNN [25] and PanNet [28] are also used for comparison. It should be noted that the TA-PNN [25] model is fine-tuned 50 epochs on the new data sets, while DML-GMME [23], PanNet [28] and DCF-4 models are not fine-tuned in this experiment. First of all, we tested the performance of models on QuickBird images. Figs. 14 and 15 show the source images and the fusion results obtained by DML-GMME [23], TA-PNN [25], PanNet [28], and DCF methods at the reduced and the full resolutions. Corresponding evaluation indexes are demonstrated in Table VIII. We can observe from Fig. 14 that the results of DML-GMME [23] and PanNet [28] suffer from serious spectral distortion in the areas of vegetations. Although TA-PNN [25] preserves the spectral information well, the fusion product of it is blurry, especially in the amplified area. Compared with these methods, DCF model makes a balance between spectral and spatial resolutions, and has a better fusion effect. In the full resolution case, it can be observed from the amplified areas in Fig. 15 that the results of DML-GMME [23] and PanNet [28] are blurry, at the same time, the strange color also appears in the results of TA-PNN [25] and PanNet [28], which means that they are spectrally distorted. Moreover, the trained models are also tested in the GeoEye-1 data set, and the results at the reduced and the full resolutions are shown in Figs. 16 and 17. The spectral distortion appears in Figs. 16(d)–(f), especially in Figs. 16(d)–(e), where the roads present green in the amplified area. Besides, blurry effect also exists in the TA-PNN [25] result. In the full resolution experiment, we can observe from Fig. 17(c) that the result of DML-GMME [23] is blurry, and

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TABLE VII

RUNTIMES OF DIFFERENT METHODS ON 256 × 256 IMAGES (SECONDS)

Methods	GSA [12]	MTF-GLP-HPM [45]	PN-TSSC [46]	VP-LGC [21]	DML-GMME [23]	TA-PNN [25]	PanNet [28]	DCF
4-bands	0.14	0.04	5.21	6.31	5.52	0.13	0.87	1.35
8-bands	0.25	0.12	8.72	11.7	9.13	0.13	0.90	3.25



(a) (b) (c) (d) (e) (f)

Fig. 15. Visual comparison among DL-based fusion methods on QuickBird data set at the full resolution (1024 × 1024). (a) PAN. (b) EXP. (c) DML-GMME [23]. (d) TA-PNN [25]. (e) PanNet [28]. (f) DCF-4.



(a) (b) (c) (d) (e) (f) (g)

Fig. 16. Visual comparison among DL-based fusion methods on GeoEye-1 data set at the reduced resolution (256 × 256). (a) PAN. (b) EXP. (c) Reference. (d) DML-GMME [23]. (e) TA-PNN [25]. (f) PanNet [28]. (g) DCF-4.



(a) (b) (c) (d) (e) (f)

Fig. 17. Visual comparison among DL-based fusion methods on GeoEye-1 data set at the full resolution (1024 × 1024). (a) PAN. (b) EXP. (c) DML-GMME [23]. (d) TA-PNN [25]. (e) PanNet [28]. (f) DCF-4.

TABLE VIII

EVALUATION INDEXES OF DL-BASED FUSION METHODS

Satellites	Methods	With Reference			Without Reference		
		Q4	SAM	ERGAS	D_λ	D_s	QNR
QuickBird	DML-GMME [23]	0.9410	3.0516	2.4507	0.0357	0.0219	0.9432
	TA-PNN [25]	0.9346	3.4285	2.4169	0.0189	0.0314	0.9503
	PanNet [28]	0.9580	3.1685	2.4464	0.0254	0.0202	0.9549
	DCF-4	**0.9725**	**2.9320**	**2.3762**	**0.0177**	**0.0195**	**0.9631**
GeoEye-1	DML-GMME [23]	0.9230	**3.2723**	2.3233	0.0363	0.0494	0.9161
	TA-PNN [25]	0.9364	3.5665	2.2396	0.0204	0.0283	0.9519
	PanNet [28]	0.9573	3.3202	2.3324	0.0235	0.0334	0.9439
	DCF-4	**0.9622**	3.3015	**2.2053**	**0.0184**	**0.0257**	**0.9564**

the aliasing effect appears in the result of TA-PNN [25]. At the same time, from the perspective of spectral resolution, the results of TA-PNN [25], and PanNet [28] are also seriously distorted. Therefore, in terms of visual inspection, DCF model performs better than other referred DL-based fusion methods on these two data sets. Furthermore, we also provide the evaluation indexes in Table VIII, where the best values are shown in bold and the second-best values are

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underlined. After inspecting Table VIII, we draw the same conclusion that although the model is trained on WV-4 data set, its generalization ability to the data sets of other satellites is also satisfying.

V. Conclusion

In this article, a DCF model has been proposed for MS and PAN image fusion. The model takes the spectral correlation collaboration and the spatial-spectral collaboration into consideration, and makes use of several subnetworks to conduct the image fusion task. In order to emphasize the spatial details and spectral consistency, DCF first separates the high-frequency and low-frequency components of PAN features, and only utilizes the high-frequency components to enhance the spatial resolution of MS images. The low-frequency components are transformed by the GG network to produce the guidance for the LR MS features to make each channel of them have a discriminative object. Then the SA is performed on each object to generate the HR MS features. Finally, the fused MS image can be obtained by the RNet. Experimental results on the reduced-resolution and full-resolution WV-4 data sets verify the effectiveness of the DCF model, and that on the QuickBird and GeoEye-1 data sets also confirm its generalization ability. Although DCF can obtain impressive fusion results, the main concern of it is the flexibility. Due to the fact that the operations on the MS images are conducted band-by-band, the four-band DCF model cannot deal with the fusion problem of MS images with other number of bands, which restricts the flexibility of the model. In the future work, we will further consider this problem and boost the flexibility of the fusion model.

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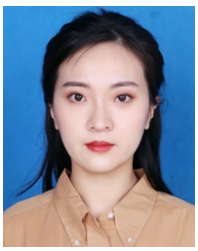
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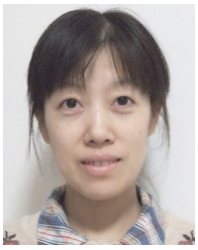
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